

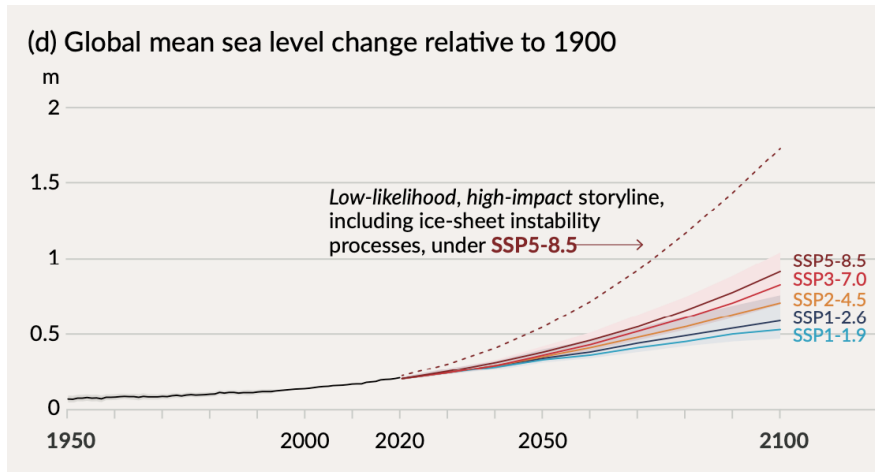


Quantifying changes in probability distributions of sea level from tide gauge data

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Student Probability Seminar
8 March 2022

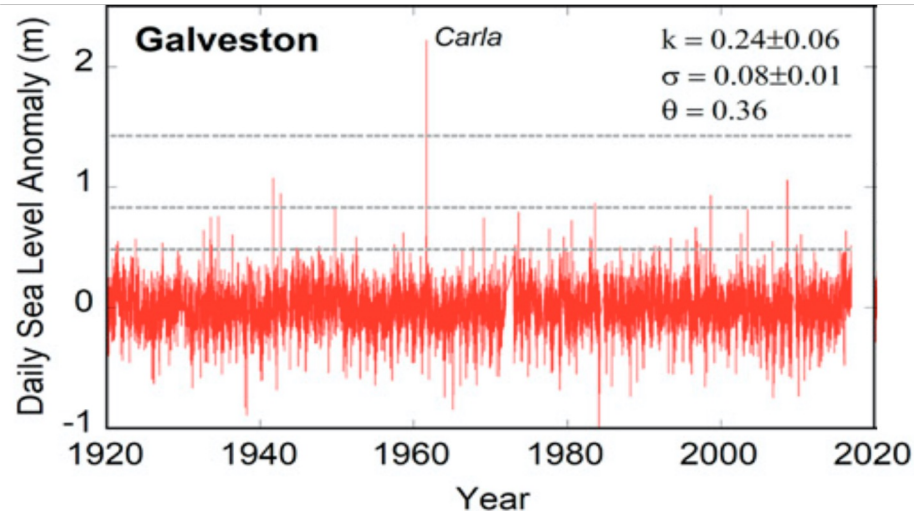
Sea level change and variability

Long-timescale changes



IPCC (2022)

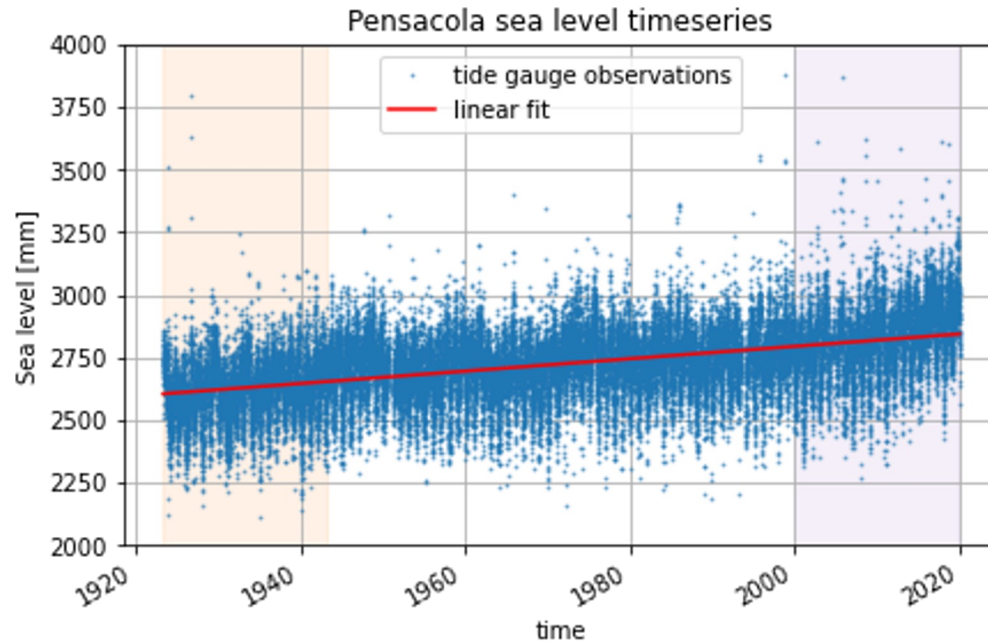
Short-timescale variability



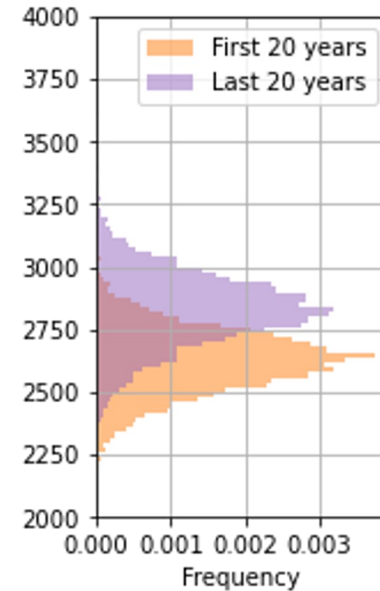
Yin et al (2020)

Predicting future sea level events

Requires understanding two components:



1. Changes in **local mean**



2. Changes in the **shape of distributions**

Research goals:

1. Quantify changes in probability distributions of observed sea level
2. Interpret changes in distribution in terms of the statistical moments

Methods: quantifying changes in sea level distributions

Extending approach of McKinnon and Rhines 2016:

- Quantile regression
- Projection onto basis functions

Observed changes in distributions

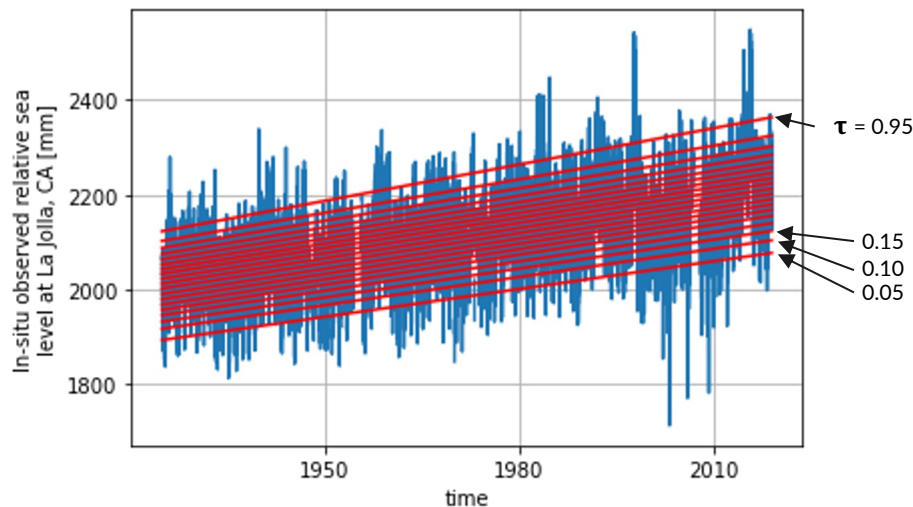
Quantile regression:

- Ordinary least squares regression seeks to estimate the conditional **mean** of a response variable Y to a predictor X :
- Quantile regression seeks to estimate conditional **quantiles** $Z(q)$ of a response variable Y to a predictor X :

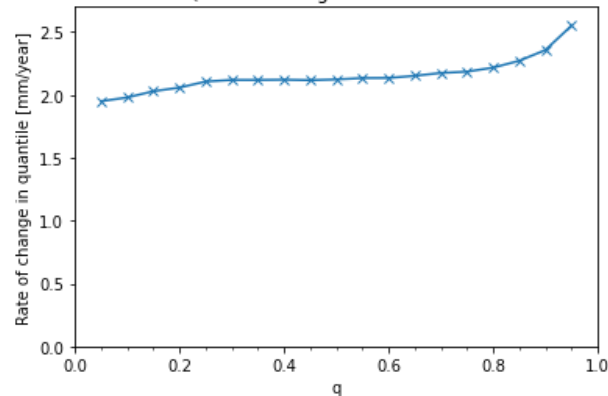
$$\mathbb{E}[Y|X] = \beta_0 + \beta_1 X$$

$$Z_{Y|X}(q) = \beta_0(q) + \beta_1(q)X$$

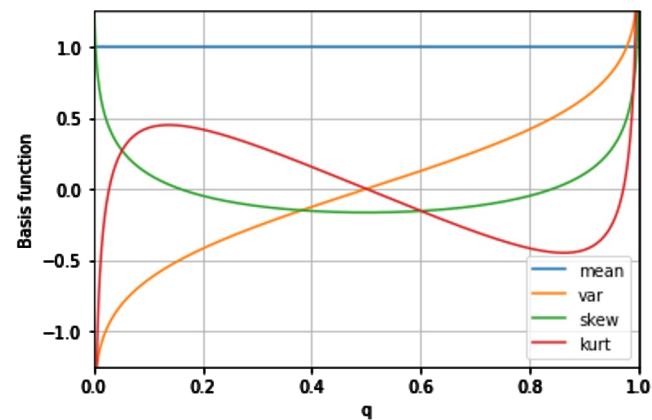
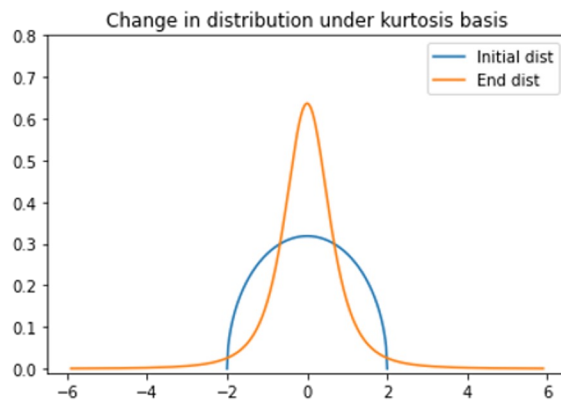
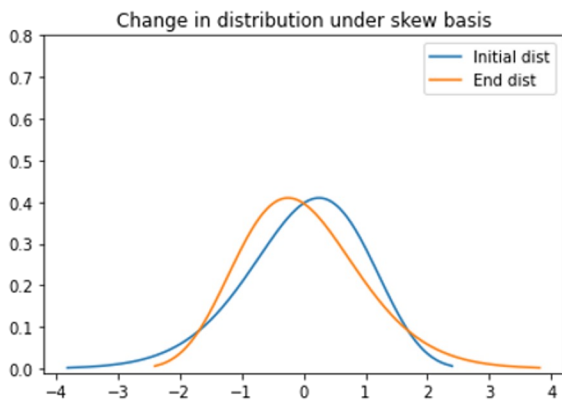
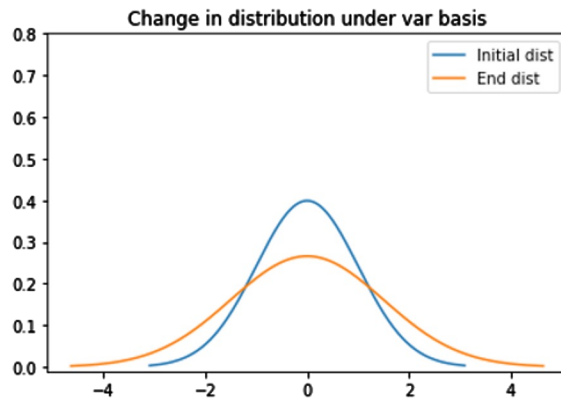
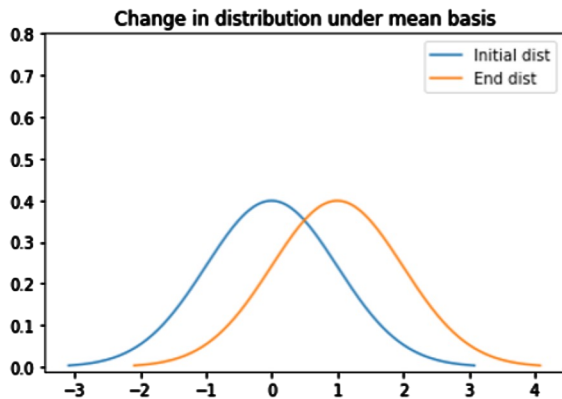
Quantile regression on a daily tide gauge timeseries



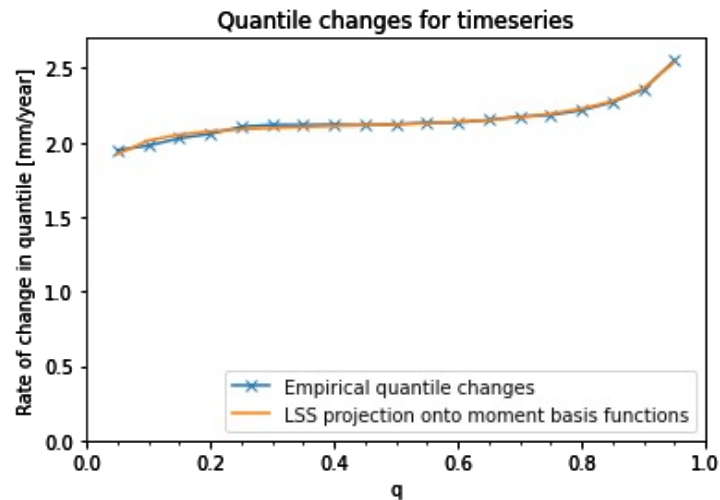
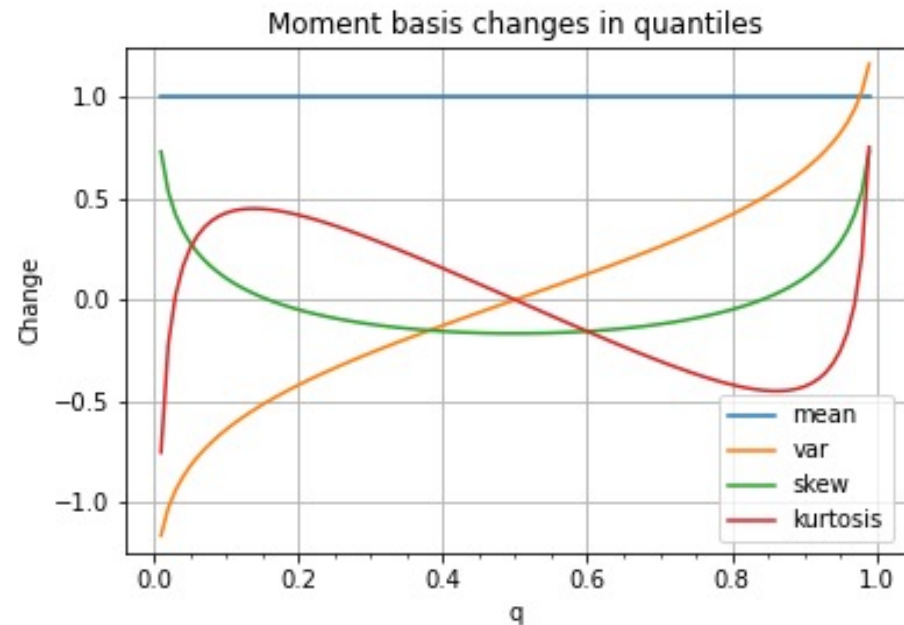
Quantile changes for timeseries



Projection onto basis functions



Projection onto basis functions



$$v \approx c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4$$

$$[c_i] = \text{mm/year}$$

$$w_i = \frac{c_i^2}{\|c\|^2}$$

weights: [0.93 0.042 0.012 0.016]

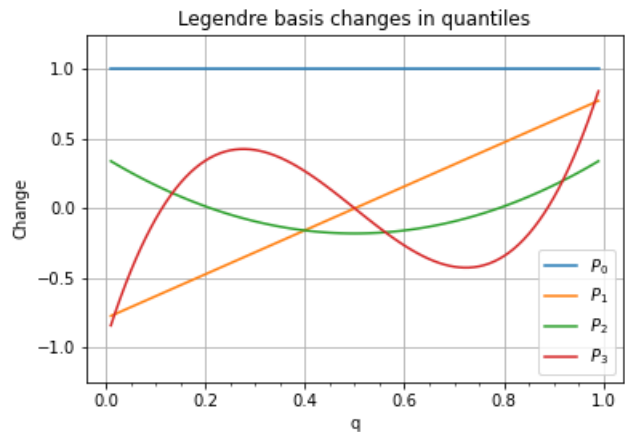
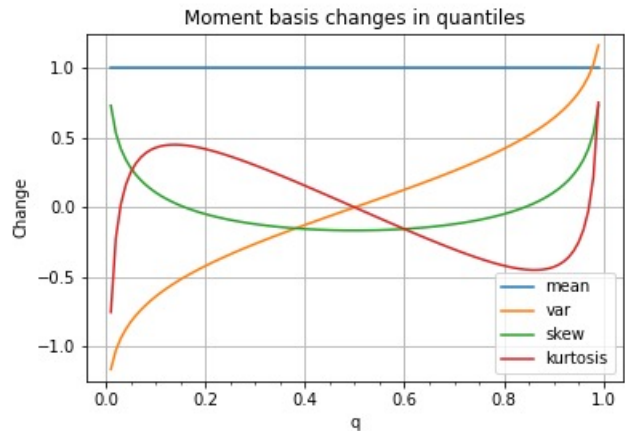
Motivation for Legendre basis

Moment basis functions are correlated:

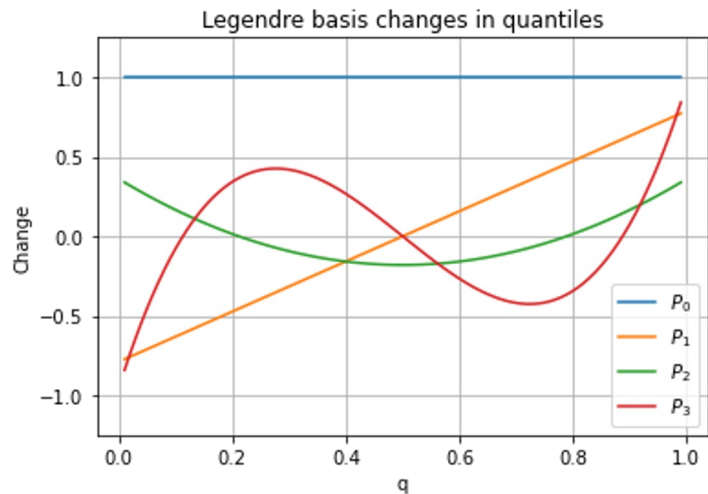
	mean	variance	skew	kurtosis
mean	1.00	-0.23	0.15	0.35
variance	-0.23	1.00	0.00	-0.90
skew	0.15	0.00	1.00	-0.00
kurtosis	0.35	-0.90	-0.00	1.00

Introduce Legendre basis:

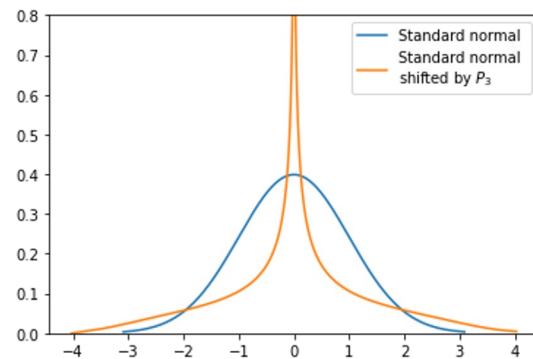
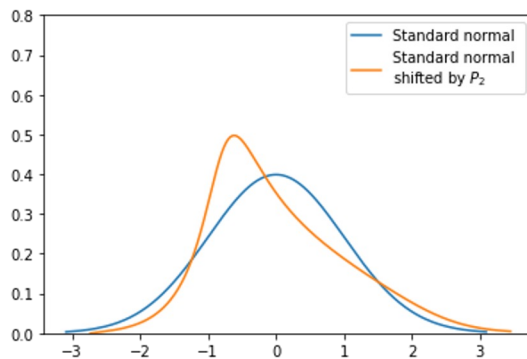
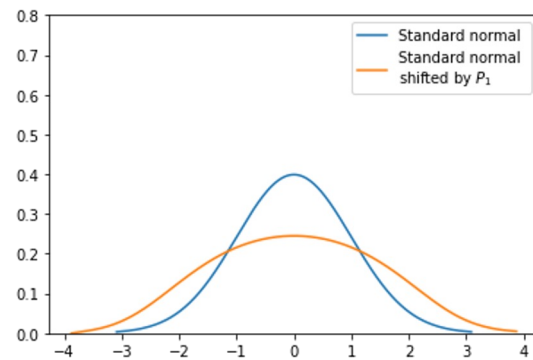
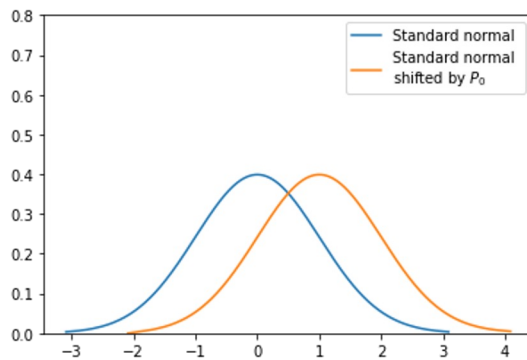
	mean	variance	skew	kurtosis
mean	1.00	0.00	-0.00	0.00
variance	0.00	1.00	0.00	-0.00
skew	-0.00	0.00	1.00	0.00
kurtosis	0.00	-0.00	0.00	1.00



Legendre basis effects on distributions



Effect of Legendre basis applied to standard normal distribution



Cornish-Fisher Expansion



- Is there a relationship between the moments of a distribution and its quantiles? Yes:
- Cornish and Fisher (1937) show that the operator $\exp\left\{\frac{a_r}{r!}\left(-\frac{d}{dx}\right)^r\right\}$ increases the r^{th} cumulant of a distribution $f(x)$, but leaves the distribution otherwise unchanged
- This allows us to estimate percentiles y_p of a distribution given the cumulants:

$$y_p \approx \mu + \sigma w;$$

$$w = z_q + (z_q^2 - 1)\frac{S}{6} + (z_q^3 - 3z_q)\frac{K}{24} - (2z_q^3 - 5z_q)\frac{S^2}{36}$$

where z_q is the q -quantile of the standard normal distribution.

- Here S and K are parameters which approximate the skewness and kurtosis for small deviations from Gaussianity.

Changes in percentiles from changes in moments



$$y_p \approx \mu + \sigma w;$$

$$w = z_q + (z_q^2 - 1)\frac{\gamma}{6} + (z_q^3 - 3z_q)\frac{\kappa}{24} - (2z_q^3 - 5z_q)\frac{\gamma^2}{36}$$

Take derivatives and evaluate near standard normal $\mathbf{k} = (\mu, \sigma^2, \gamma, \kappa) = (0, 1, 0, 0)$:

$$\left. \frac{\partial y_p}{\partial \mu} \right|_{\mathbf{k}} = 1$$

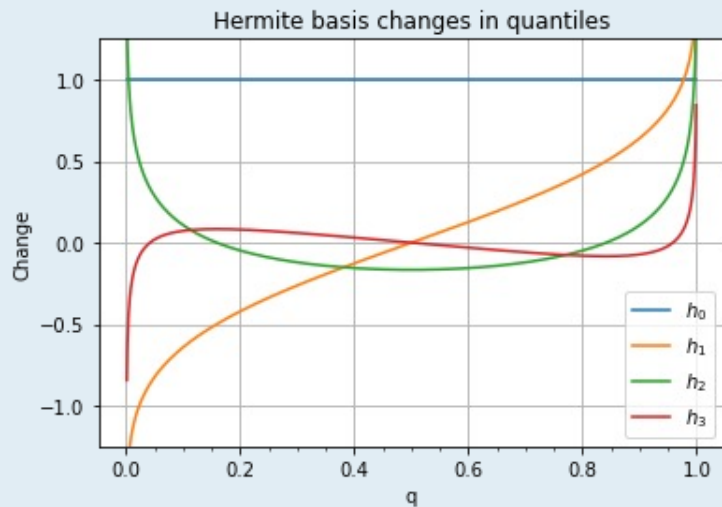
$$\left. \frac{\partial y_p}{\partial \sigma^2} \right|_{\mathbf{k}} = \frac{1}{2} z_q$$

$$\left. \frac{\partial y_p}{\partial \mu} \right|_{\mathbf{k}} = \frac{1}{6} (z_q^2 - 1)$$

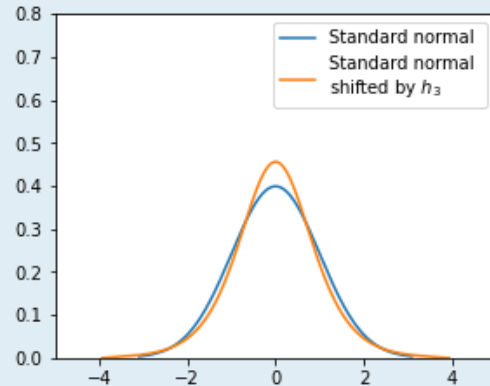
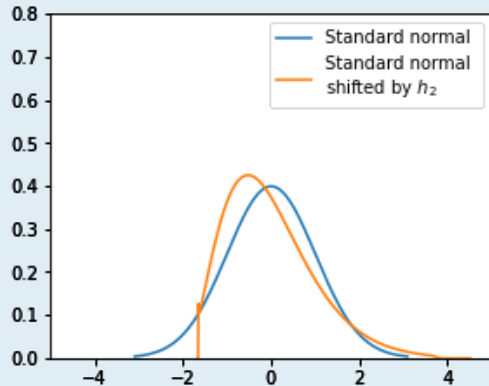
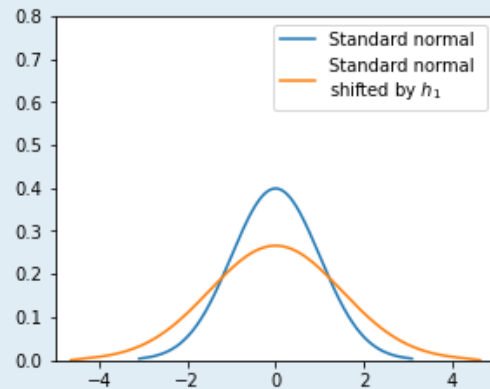
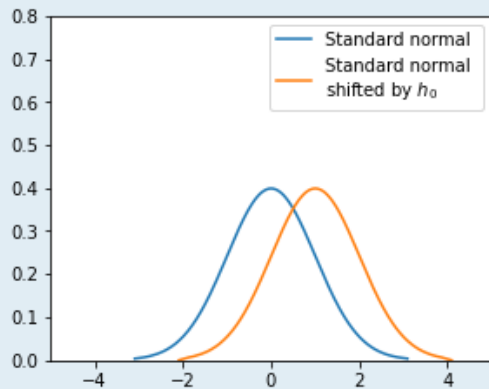
$$\left. \frac{\partial y_p}{\partial \mu} \right|_{\mathbf{k}} = \frac{1}{24} (z_q^3 - 3z_q)$$

These are Hermite polynomials of z_q – **and form an orthogonal set of functions** on $L^2[0,1]$

Hermite basis effects on distributions



Effect of Legendre basis applied to standard normal distribution

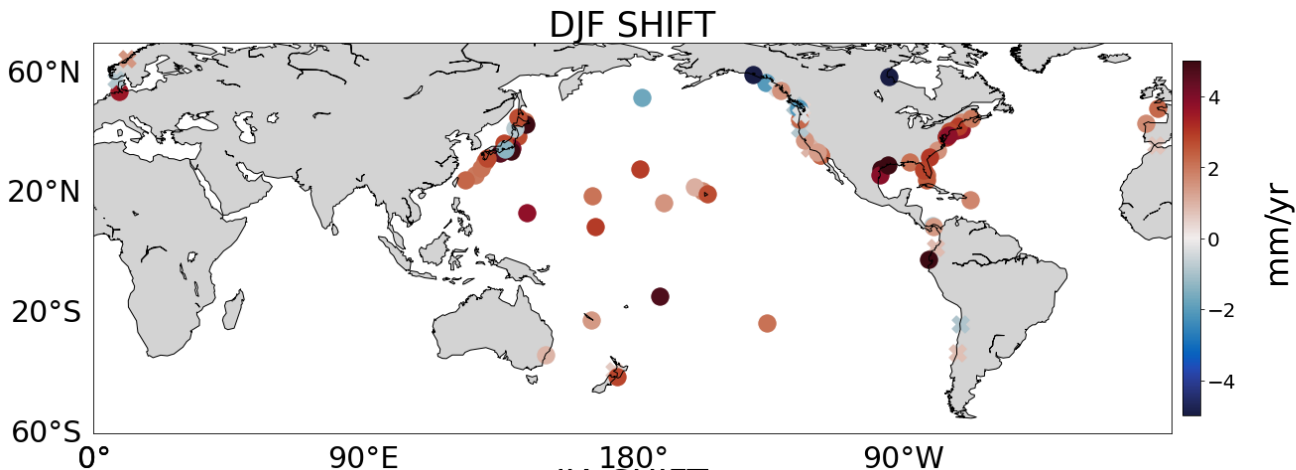


Results: application to tide gauges

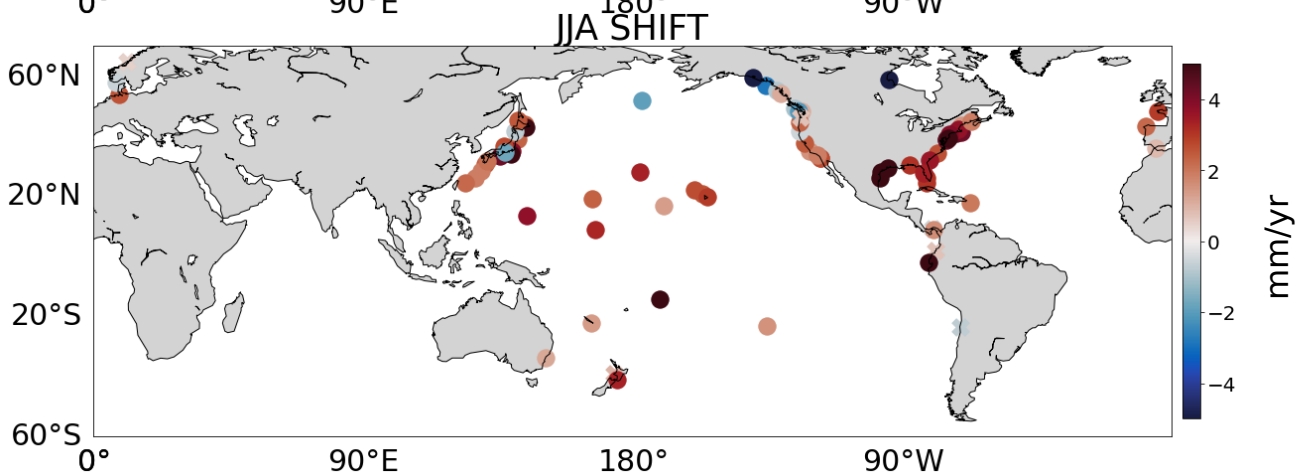
- Daily sea level from 80 tide gauges from 1970-2018
- Divide into winter and summer seasons (DJF and JJA)
- Statistical significance via block bootstrapping

Changes in mean sea level

- = statistically significant
- ✕ = lacks significance



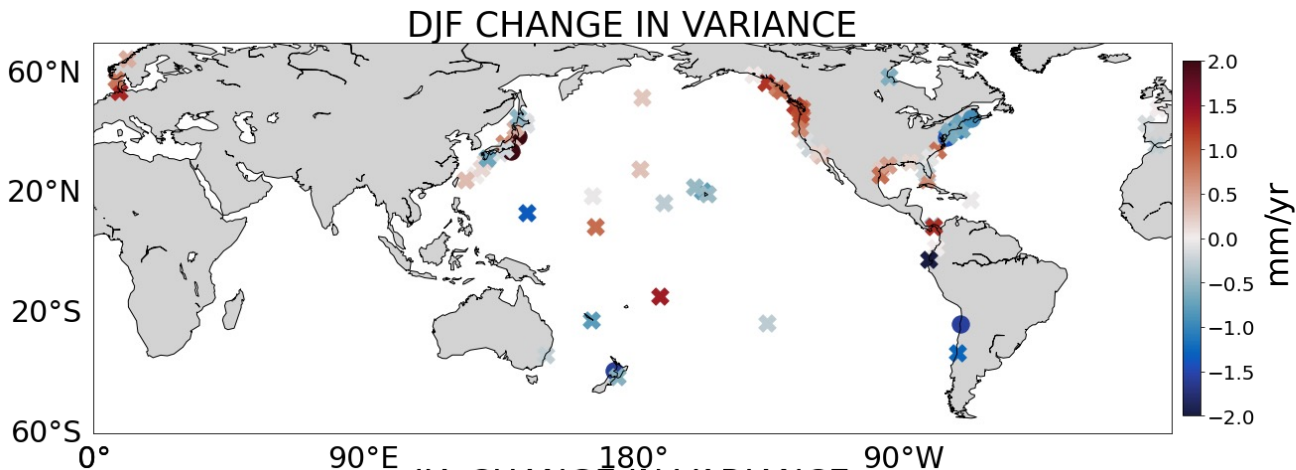
Statistical significance for
67/80 tide gauges



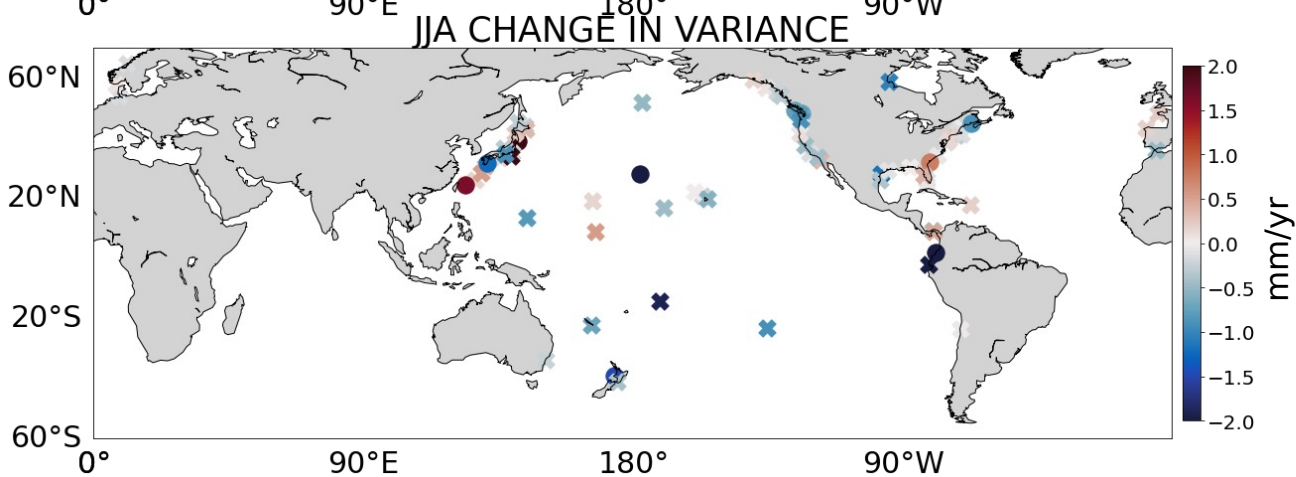
Statistical significance for
73/79 tide gauges

Changes in sea level **variance**

- = statistically significant
- ✕ = lacks significance



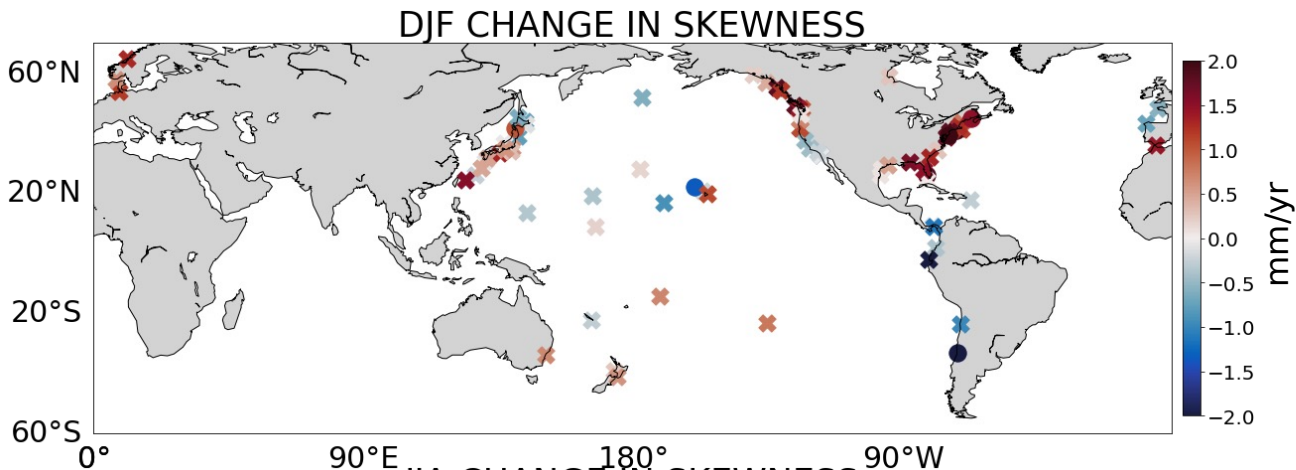
Statistical significance for
6/80 tide gauges



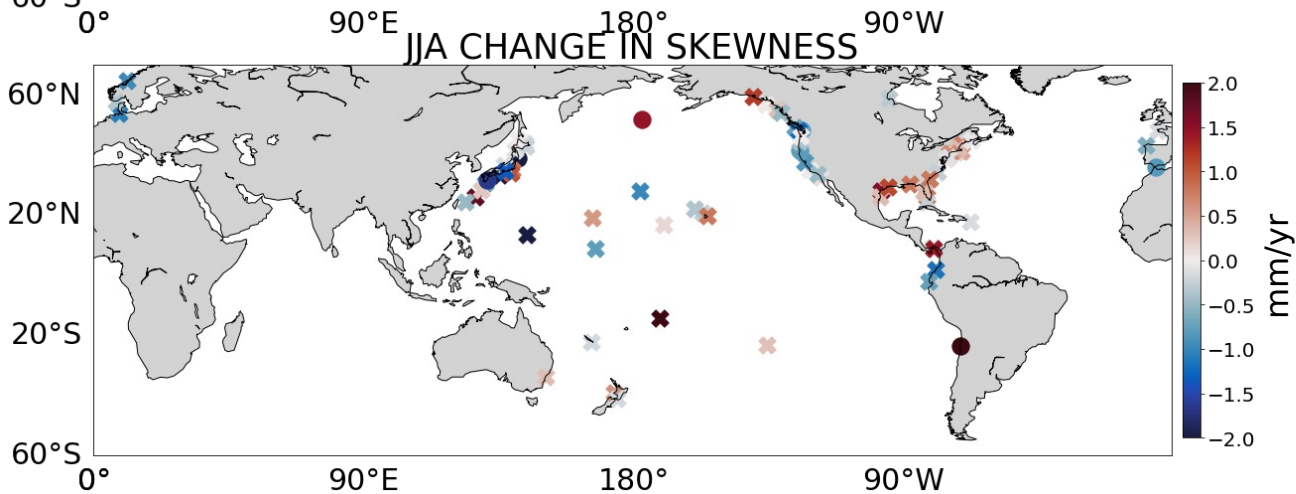
Statistical significance for
10/79 tide gauges

Changes in sea level skewness

- = statistically significant
- ✕ = lacks significance



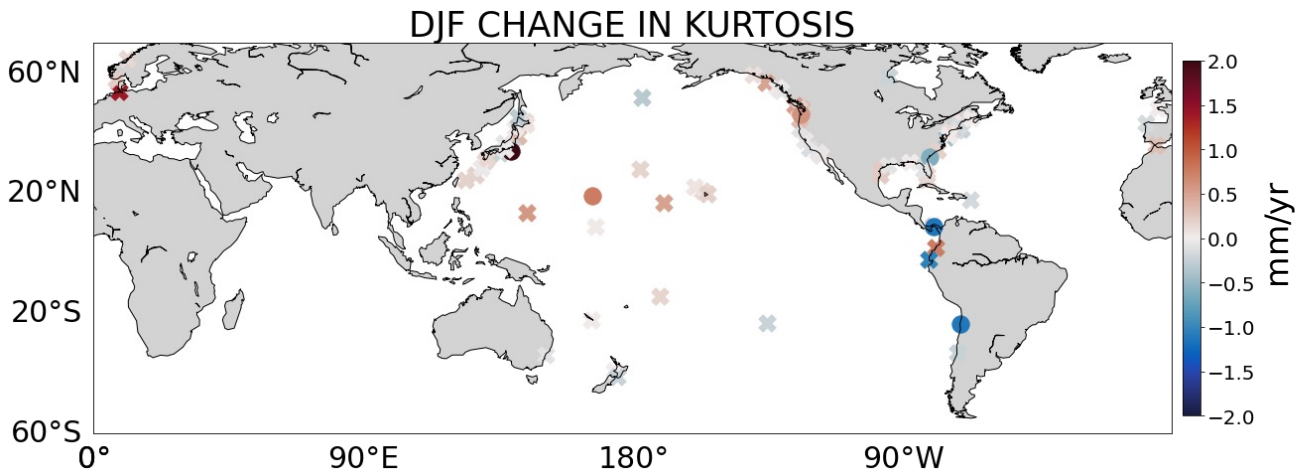
Statistical significance for
9/80 tide gauges



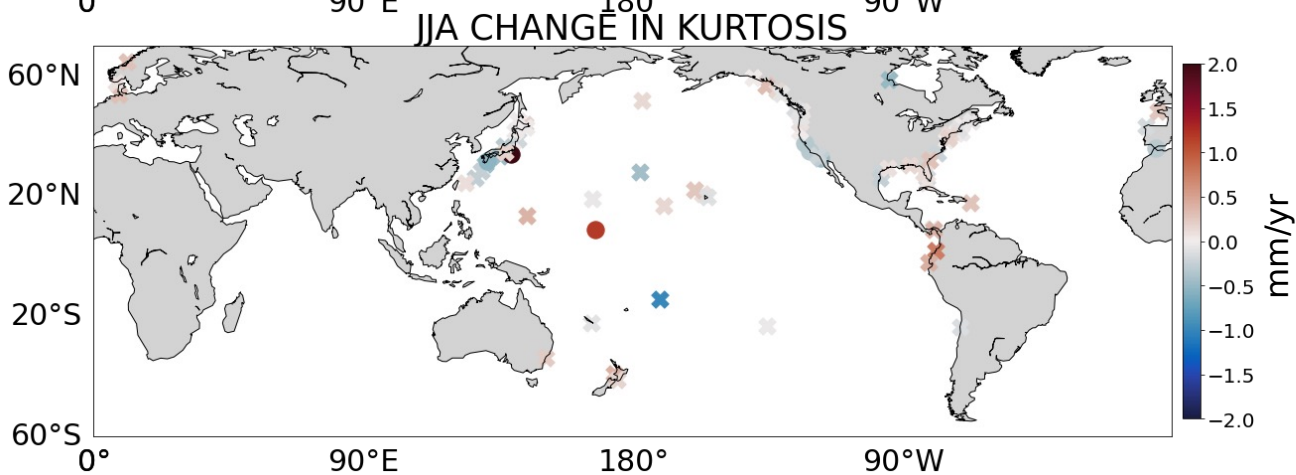
Statistical significance for
7/79 tide gauges

Changes in sea level **kurtosis**

- = statistically significant
- ✕ = lacks significance

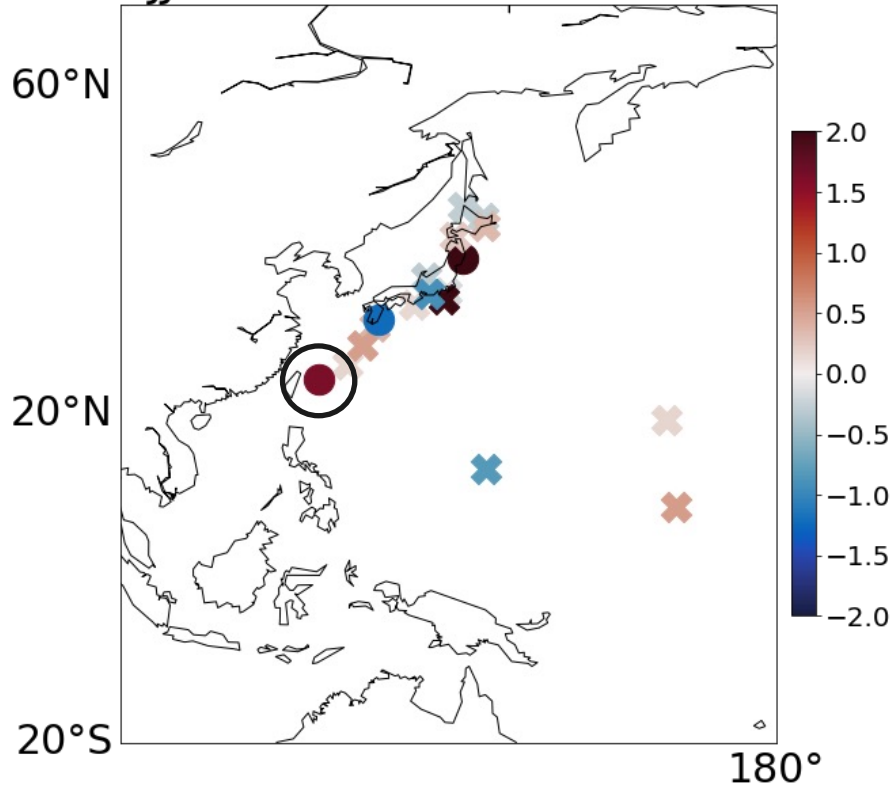


Statistical significance for
6/80 tide gauges

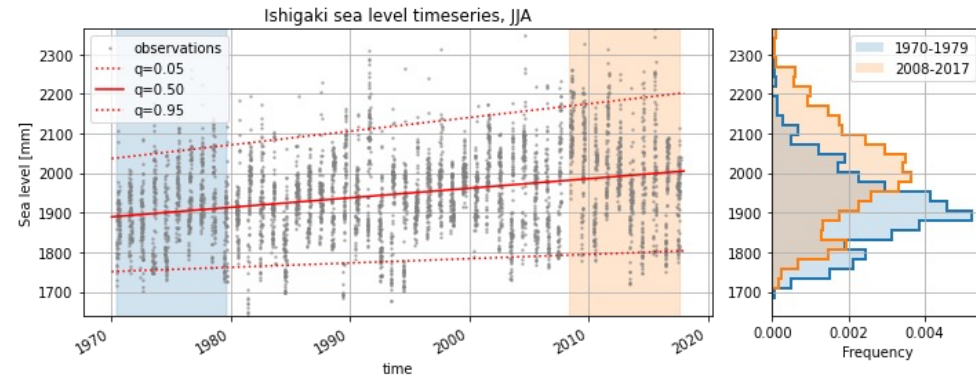


Statistical significance for
7/79 tide gauges

JJA CHANGE IN VARIANCE



Example tide gauge with statistically significant changes in mean and variance



Assessment to 2100 of the effects of reef formation on increased wave heights due to intensified tropical cyclones and sea level rise at Ishigaki Island, Okinawa, Japan

Chuki Hongo ^{a,b} and Masashi Kiguchi ^c



Conclusions

1. Dominant change in sea level distributions is due to changes in the mean, consistent with previous studies
2. Some tide gauges show statistically significant changes in higher order moments as well, indicating that changing probability distributions can be important



References

1. IPCC, 2021: Summary for Policymakers. In: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson- Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)]. Cambridge University Press.
2. Kopp RE, Horton RM, Little CM, Mitrovica JX, Oppenheimer M, Rasmussen D, et al. Probabilistic 21st and 22nd century sea-level projections at a global network of tide-gauge sites. *Earth's future*. 2014;2(8):383–406.
3. Yin J, Griffies SM, Winton M, Zhao M, Zanna L. Response of Storm-Related Extreme Sea Level along the US Atlantic Coast to Combined Weather and Climate Forcing. *Journal of Climate*. 2020;33(9):3745–3769.
4. McKinnon KA, Rhines A, Tingley MP, Huybers P. The changing shape of Northern Hemisphere summer temperature distributions. *Journal of Geophysical Research: Atmospheres*. 2016;121(15):8849–8868.
5. Woodworth PL, Blackman DL. Evidence for systematic changes in extreme high waters since the mid-1970s. *Journal of Climate*. 2004;17(6):1190–1197.